Sukhrob Ilyosbekov

Computer Vision · Deep Learning · Explainable AI

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Research Interests

My research is in computer vision for medicine and biology. I'm most interested in models whose decisions an expert can check, and in representations that carry over to images the model wasn't trained on. I'd like to spend a PhD on these problems.

Education

Northeastern University, Boston, MA *January 2024 – May 2026*

- Master of Science in Computer Science; GPA 4.0/4.0
- Coursework: Advanced Perception, Computer Vision, Machine Learning

Suffolk University, Boston, MA *September 2021 – May 2023*

- Bachelor of Science in Computer Science; Cum Laude; GPA 3.5/4.0
- Transferred from INTI International College, Malaysia (2019 – 2021) and Tashkent University of Information Technologies, Uzbekistan (2017 – 2019)

Publications

[1] **S. Ilyosbekov**, S. Gajjar, and R. Jin, "MorphoCLIP: Text-Supervised Contrastive Learning for Perturbation Matching in Cell Painting Images," *arXiv preprint*, arXiv:2608.22690, 2026. [Paper]

[2] **S. Ilyosbekov**, "Localize, Don't Beautify: Client-Side Control of Image-Editing APIs for Cosmetic Surgery Previews," *arXiv preprint*, arXiv:2608.02841, 2026. [Paper]

[3] **S. Ilyosbekov**, "MelanomaNet: Explainable Deep Learning for Multi-Class Skin Lesion Classification," *arXiv preprint*, arXiv:2512.09289, 2025. [Paper]

Awards & Honors

- **Associate Member,** Sigma Xi, The Scientific Research Honor Society, 2026
- **Outstanding Project Award,** Bookshelf Scanner, Northeastern University
- **National Physics Olympiad, 1st Place,** Uzbekistan; first student from my high school to win nationally
- **University Coding Competition Winner,** Tashkent University of Information Technologies

Research

MorphoCLIP: Text-Supervised Cell Painting Retrieval GitHub | Paper

PyTorch, DINOv3, BioClinical ModernBERT, contrastive learning, CPJUMP1

- Learns a shared space for Cell Painting images and text, so a compound or gene perturbation can be retrieved from a written description of it, and the other way round.
- DINOv3 and BioClinical ModernBERT stay frozen and only the projection heads train, which fits on a single consumer GPU.
- Evaluated on CPJUMP1: 51 plates, 3M+ images, 303 compounds, and 160 genes. Batch correction in embedding space keeps the model from simply recognizing plates.
- First author, with S. Gajjar and R. Jin. I led model design, training, and evaluation.

Localize, Don't Beautify: Client-Side Control of Image-Editing APIs Paper

Image-editing APIs, inpainting models, ArcFace, prompt engineering

- Studies how far a client can control a black-box editing API, using cosmetic-surgery previews as the test case.
- I ran prompt steering, masked compositing, and model-based inpainting through six commercial editors on 196 facelift and rhinoplasty edits.
- Masked compositing, the simplest of the three, kept edits localized best, with identity preservation scored by ArcFace. Sole author.

MelanomaNet: Explainable Deep Learning for Skin Lesion Classification GitHub | Paper
PyTorch, EfficientNet V2, focal loss, GradCAM++, OpenCV, ISIC 2019

- EfficientNet V2 (384×384, focal loss) over all nine ISIC 2019 classes, reaching 0.86 weighted F1 on 25K dermoscopic images.
- Splits GradCAM++ attention along the ABCDE criteria. Asymmetry, border irregularity, color variation (K-means), and diameter come straight from the lesion mask, and the paper scores how closely the model's attention agrees with each one. Sole author.

Selected Projects

LightDepth: Lightweight Monocular Depth Estimation GitHub
PyTorch, ResNet18, U-Net, NYU Depth V2

- ResNet18 encoder and U-Net decoder for single-image depth. It has 42% fewer parameters than Depth Anything V2 (14.3M vs. 24.8M) and runs 72% faster.
- Error on NYU Depth V2 is lower too: AbsRel down 5.3% and SqRel down 31.1% against the same baseline.

FSRCNN: Accelerating Super-Resolution Convolutional Neural Network GitHub
PyTorch, mixed-precision training, T91, Set5, Set14, DIV2K

- Reimplemented FSRCNN (Dong et al., ECCV 2016) at 2×, 3×, and 4×, with upsampling learned end to end.
- Recovered the paper's improvement over SRCNN (+1.78 dB PSNR on Set5, +1.26 dB on Set14), then ran ablations on the shrinking and mapping layers.

Bookshelf Scanner: Multi-Modal Book Detection and Recognition GitHub
YOLO, Moondream2 VLM, llama.cpp, PyTorch, FastAPI

- YOLO instance segmentation finds each spine in a shelf photo, and Moondream2 reads the title and author.

Work Experience

Generative AI Engineer | Sidebench | Los Angeles, CA *Jun 2026 – Present*

- Build proofs of concept that test off-the-shelf image and language models on a client's task before it is built, including cosmetic surgery previews and illustrated social stories for autistic children.
- Evaluation covers identity preservation (ArcFace), landmark-masked edits, automated content checks and human rubrics. The cosmetic surgery work led to the *Localize, Don't Beautify* preprint.

Machine Learning Engineer | EmTech Care Labs | Portland, ME *Jan 2025 – Jun 2025*

- Built a model that uses daily activity and caregiver logs to flag Alzheimer's residents at risk of decline, so staff can step in earlier.
- Wrote an NLP pipeline that turns free-text clinical notes into structured records of symptoms, medications, and care events. Both models run in production on de-identified data under HIPAA.

Machine Learning Engineer | AllFactors | Santa Clara, CA *Oct 2021 – Dec 2024*

- Built the property-valuation and market-forecast models that replaced the company's rules-based system.
- Wrote the real-time feature pipeline that fed them (SignalR, SQL Server change tracking) and the Blazor dashboards where users saw the results.

Earlier: Software Engineer, Smart Meal Service, Russia (2020 – 2021). Game Developer, Pentalight Technology, Malaysia (2020 – 2021). Freelance (2019 – 2020). Game Developer, EC Dev Team, Uzbekistan (2016 – 2019).

Technical Skills

Deep Learning & CV: PyTorch, TensorFlow/Keras, OpenCV, CNNs, Vision Transformers, DINOv3, U-Net, EfficientNet, YOLO, contrastive and self-supervised learning, GradCAM++, detection and segmentation, depth estimation

Mathematics: Linear algebra, probability and statistics, convex optimization, numerical methods, information theory, hypothesis testing and experimental design

Machine Learning: Transfer learning, loss design, regularization, hyperparameter search, cross-validation and error analysis, XGBoost, K-means, NLP

Research Computing: Python, NumPy/SciPy, pandas, scikit-learn, Hugging Face, CUDA, mixed-precision and multi-GPU training, Weights & Biases, Docker, LaTeX

Software Engineering: C#, C++, TypeScript; .NET, FastAPI, PostgreSQL, Kubernetes, AWS, Azure